Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 17

Equivalence Classes and Partitions

Covers Chartrand & Zhang, Section 5.2

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Compute one class, then find one you missed

Finding all the distinct classes

1. Pick any element a and compute $[a] = \{x : x \, R \, a\}$.
2. Pick an element *not* in any class found so far, and compute its class.
3. Repeat until every element is accounted for.

By Theorem 5.23 the classes you produce are disjoint and cover the set, so this terminates and misses nothing.

The two-way correspondence

Equivalence relation → partition: the distinct classes are the blocks.
Partition → equivalence relation: relate a and b when they share a block.
If you are given either one, you can produce the other.

Three mistakes to expect

1. **Listing $[a]$ for every a and calling them all distinct.** Different labels often name the same class.
2. **Producing overlapping classes.** If two of your classes share an element they are the same class — recheck the relation.
3. **Forgetting the diagonal** when reconstructing R from a partition. Every (a, a) must be there.

Part II: Practice Problems

Section A — Computing classes

1. Let $S = \{1, 2, 3, 4, 5, 6, 7\}$ and let R be the equivalence relation

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,7), (1,5), (5,1), (2,4), (4,2), (2,7), (7,2), (4,7), (7,4)\}.$$

Determine the distinct equivalence classes.

2. For each equivalence relation on $\mathbb{Z}$, describe the distinct equivalence classes.

(a) $a \mathrel{R} b$ if $a - b$ is divisible by 4.

(b) $a \mathrel{R} b$ if $a^2 = b^2$.

(c) $a \mathrel{R} b$ if $a = b$.

Section B — Going backwards

3. Let $S = \{a, b, c, d, e, f\}$. If the distinct equivalence classes of R are $\{a, d\}$, $\{b, f\}$, and $\{c, e\}$, write out R and state how many ordered pairs it contains.

4. An equivalence relation on $S = \{1, 2, 3, 4, 5, 6\}$ has exactly three distinct classes, and you are told $3 \in [4]$, $[2] \cap [6] = \emptyset$, and $1 \in [3]$. Determine the classes.

Section C — The theorem

5. Explain, in two or three sentences each, which property of an equivalence relation guarantees:

(a) that no equivalence class is empty;

(b) that two classes sharing an element must be equal.

Section D — Writing

6. In five to eight complete sentences, explain why an equivalence relation and a partition are two descriptions of the same thing. Give a concrete example from computing, and say what plays the role of each.

Section E — Challenge (optional)

7. How many different equivalence relations are there on a set of three elements? List the corresponding partitions.